MERIDIAN TRANSIT SURVEY · RESEARCH NOTE 7

A temperate sub-Neptune around a nearby red dwarf

Nine transits of MTS-0412 b in 118 nights of photometry



MTS-0412 b in three numbers

2.2

times the radius of the Earth

11.8

days to complete one orbit

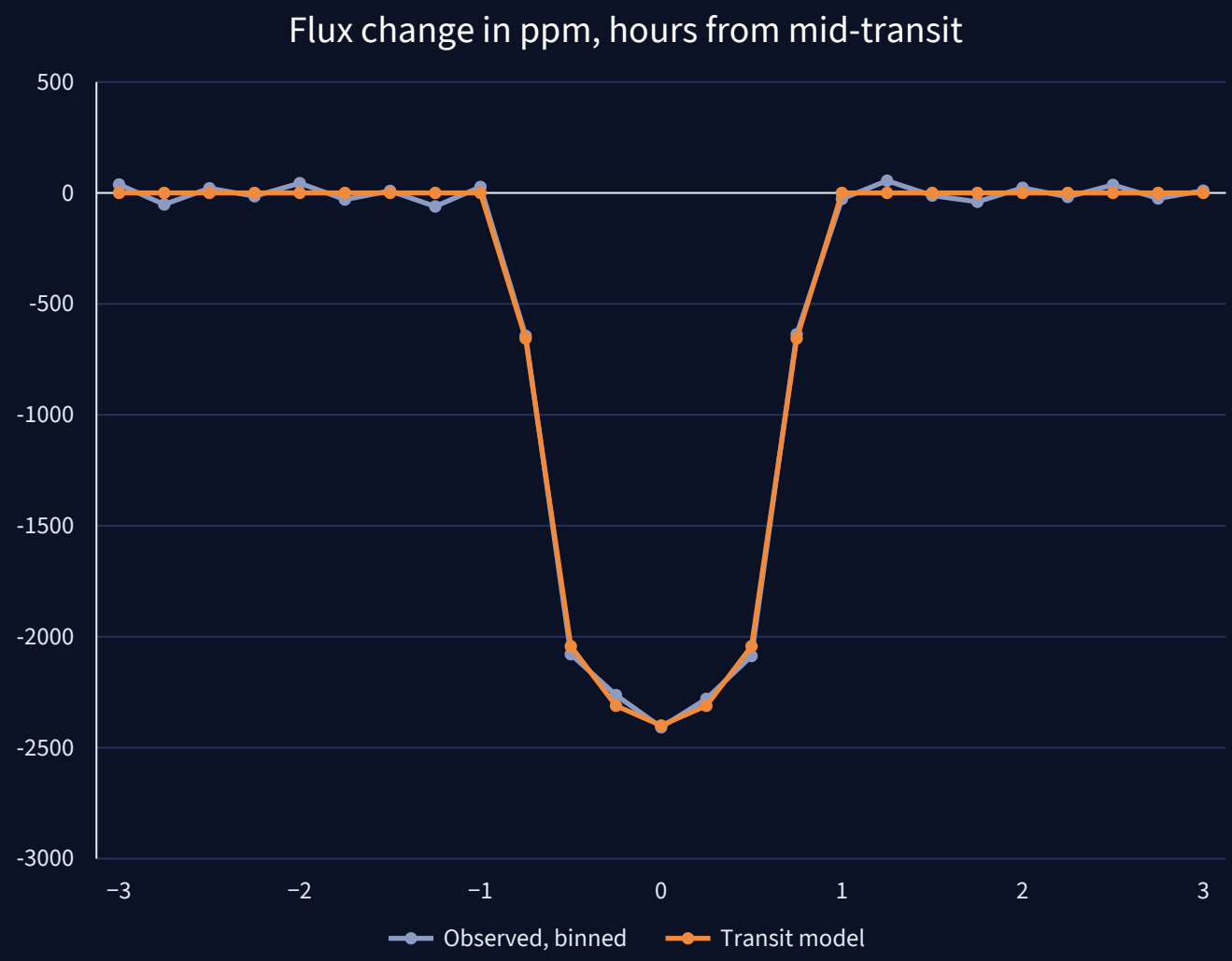
1.3×

the sunlight that reaches the Earth

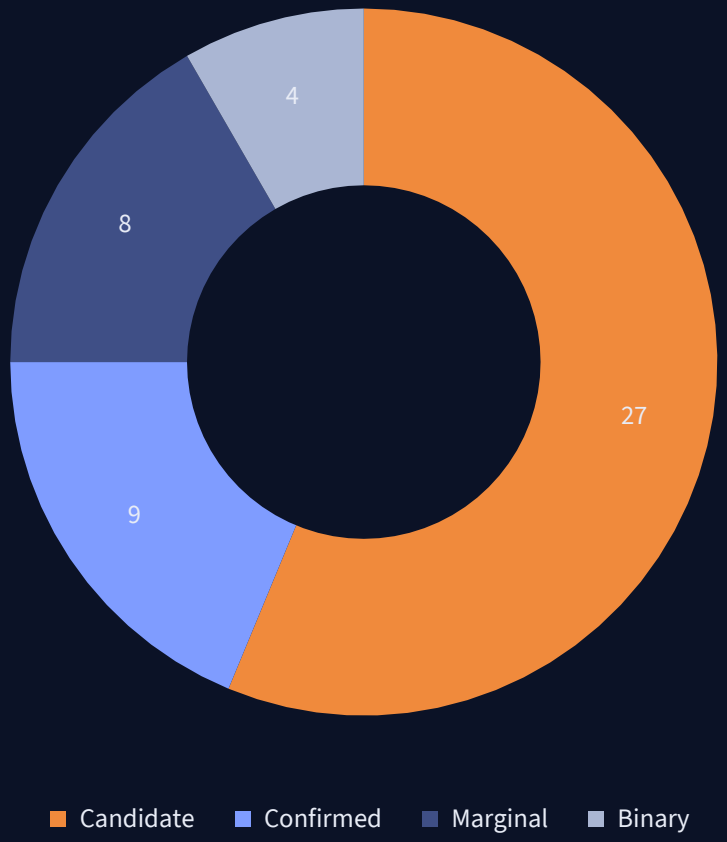
A dip of a quarter of one percent

Every 11.82 days the star dims by 2,400 parts per million for 1.7 hours.

The flat floor of the dip rules out a grazing eclipse between two stars.



48 candidates, and what comes next



Radial velocities · spring 2027

↓

Mass and density · summer 2027

↓

Transmission spectrum proposal